LEAN SIX SIGMA DMAIC PROJECT

Improving Energy Efficiency and Operational Reliability of Data Centers

An industry-level DMAIC case built only on published, sourced data

Project Information	Details
Subject	Data centers (enterprise, colocation and hyperscale) — facility energy, availability and water
Methodology	Lean Six Sigma — DMAIC
Reference process	The facility-management process of a typical data-center operator whose performance equals the published industry baseline (Uptime Institute 2026)
CTQs	4: PUE (energy efficiency); impactful-outage incidence; outage preventability; water-use tracking
Project cycle	12 weeks planned (Define 2, Measure 2, Analyze 3, Improve 3, Control 2)
Data policy	**Real data only.** Every number is taken from a cited public source or is arithmetic on those numbers; every source is hyperlinked. No simulated, estimated-by-the-author or fictional data.
Data cut-off	Published sources available as of 30 September 2026 (latest: Uptime Global Data Center Survey 2026, released 28 July 2026)

How to read the source tags. Superscript tags such as [1] are clickable and open the source; the full list is in the References section. Calculations made by the author from source numbers are marked “computed”.

Executive Summary

This project applies Lean Six Sigma DMAIC to the operation of data centers, an industry whose electricity use is rising quickly: U.S. data centers used 58 TWh in 2014 and 176 TWh in 2023 (4.4% of U.S. electricity), and 325–580 TWh in total is projected for 2028 [5] [12]. The study defines four Critical-to-Quality (CTQ) requirements — power usage effectiveness (PUE), impactful-outage incidence, outage preventability, and water-use tracking — and measures them against the latest published industry benchmark, the Uptime Institute Global Data Center Survey 2026 [1].

Baseline. Industry-average PUE is 1.52 (n=644): overhead of 0.52 kWh per kWh of IT load, and only 23% of respondents operate a data center below 1.3 [1]. Forty-seven percent of operators had an impactful outage in the past three years (n=715), 71% of those with a significant outage say it cost at least $100,000, and 89% judge their most recent impactful outage avoidable with better management, processes or configuration [1]. Only 53% collect water-use data [1].

Analysis. A control chart of eight years of published PUE shows a *stable, slowly improving* process (2019–2026 limits 1.493–1.652; slope -0.0093 per year), which at that pace would need about 13 years to reach PUE 1.40 (computed). Outage incidence fell from 74% to 47% since 2020 ($z = 9.4$, $p<0.001$, computed), but the latest one-year change (50%→47%) is not statistically significant ($p = 0.25$). Root causes verified from published evidence are: exhausted easy efficiency gains and long-lived legacy sites; power-chain failures (UPS 39%, transfer switch 34%, generator 28% of power-related outages) combined with ignored or flawed procedures (85% of human-error outages) [1] [3]; and water sitting outside the standard KPI set.

Improve and Control. Countermeasures are tied to each verified cause and backed by published results — for example machine-learning cooling control (−40% cooling energy, −15% PUE overhead) [7] and hyperscale fleets operating at PUE 1.09 [6]. Because no pilot was run at a real facility, results are presented as *published evidence and transparent projections*, not as measured pilot gains. Applying the DeepMind result to the industry baseline would give PUE 1.442; reaching 1.40 needs a 23% overhead reduction (computed). The Control plan sets standards, charts, owners, action limits and response plans.

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1. Industry Profile and Project Context

1.1 Industry and scope

A data center houses IT equipment and the power and cooling systems that keep it running. Operators are enterprises running their own facilities, colocation providers, and hyperscale cloud companies. In 2026, for the first time, third-party facilities and services hold a larger share of corporate IT workloads (46%) than enterprise-owned data centers (44%) [1]. The Uptime survey of more than 800 owners and operators is the longest-running study of its kind (16th edition) [1] [10].

The selected improvement area is the **facility-management process**: the chain from utility feed, through power conditioning and backup, to the IT load, and the removal of the resulting heat, together with the metering, incident logging and reporting that describe performance. Building construction, IT application design, server procurement and market pricing are out of scope.

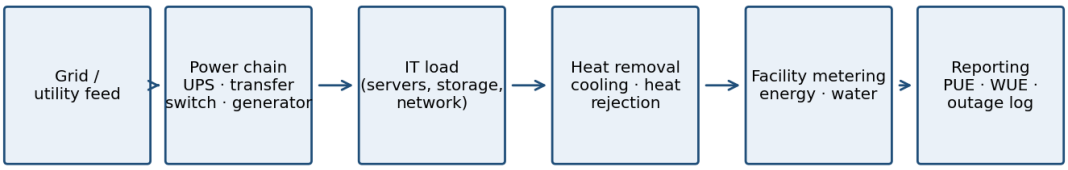


Figure 1. Facility-management process included in the project scope (schematic; no data shown).

1.2 CTQ ownership (proposed design)

Process area	Suggested process-owner role	Primary CTQ
Power and cooling engineering	Facility Engineering Manager	CTQ 1 — PUE
Critical-facilities operations	Operations / Critical Facilities Manager	CTQ 2 — Impactful-outage incidence
Procedures, training and change management	Operational Excellence / Procedures Owner	CTQ 3 — Outage preventability
Environmental reporting	Sustainability / Energy Manager	CTQ 4 — Water-use tracking

Roles are a recommended assignment for an operator adopting this project; they are design choices, not data.

1.3 Data and integrity note

The sample project used as a model for this report relied on simulated data. This report deliberately does not: it uses only public, citable data. Consequences the reader should know: (1) the “baseline” is the published industry benchmark rather than one site’s logs; (2) survey figures are self-reported by operators and are not a random sample; (3) PUE is defined by operators for their largest site, and Google’s fleet PUE is measured under its own boundary [1] [6]; (4) no pilot was executed, so the Improve phase reports published results and clearly labelled projections; (5) FMEA severity/occurrence/detection scores were not used because they require site-specific expert judgement that cannot be sourced — risk ranking uses published frequencies instead.

2. DEFINE Phase

2.1 Project charter

Charter element	Project definition
Business case	Energy overhead, unplanned outages and water use raise cost and risk. Cost is the top management concern (79% at least somewhat concerned) [1]; one in five operators with a significant outage report a cost above $1 million [1]; data-center electricity in the U.S. rose from 58 TWh (2014) to 176 TWh (2023) [12].
Problem statement	Baseline performance (Uptime 2026) is below target on all four CTQs: PUE 1.52; 47% of operators with an impactful outage in three years; 89% of impactful outages avoidable; 53% tracking water use [1].
Goal statement	Within one 12-week improvement cycle, define and start executing a plan that brings an operator with baseline performance to: average PUE ≤ 1.40; impactful-outage incidence ≤ 35%; avoidable share of outages ≤ 75%; water-use tracking ≥ 79% of sites.
In scope	Power chain (UPS, transfer switches, generators), cooling and heat rejection, procedures and change management, energy and water metering and reporting.
Out of scope	Building construction, IT application design, server purchasing, colocation pricing, cyber-security programmes (cyber incidents are noted only as an outage cause).
Project sponsor	Head of Infrastructure / Data-center Operations (proposed).
Timeline	12 weeks: Define 2, Measure 2, Analyze 3, Improve 3, Control 2.

2.2 Voice of the stakeholder and CTQ translation

Stakeholder needs below are documented concerns from the cited sources (paraphrased); no interview quotes were invented.

Stakeholder	Documented need / concern	CTQ	Specification / project target
Operator management	Cost control: 79% at least somewhat concerned about cost; improving facility energy performance is a stated concern [1]	PUE	Individual site ≤ 1.5; project average ≤ 1.40; ≥ 35% of sites ≤ 1.3 (from 23%)
Regulators	Germany requires existing sites ≤ 1.5 (2027) and ≤ 1.3 (2030); a 2026 draft amendment proposes 1.6 and 1.4 [8] [9]	PUE	Individual ≤ 1.5 (current statutory value)
Business users / tenants	Downtime is costly: 71% of operators with a significant outage report ≥ $100,000 [1]	Outage incidence	Individual: no serious/severe outage; project ≤ 35% with any impactful outage per 3 years
Operators / auditors	Redundancy only works if configured, maintained and operated well; 89% say their outage was avoidable [1]	Preventability	≤ 75% of impactful outages judged avoidable
Communities / regulators	Water scarcity is a growing public concern; water tracking is rising under public pressure and legislation [1] [2]	Water tracking	Site reports annual water use; ≥ 79% of sites (parity with PUE tracking, 79%)

Project targets are design choices by the author, anchored to the cited benchmarks (e.g. 1.40 equals the LBNL modelled U.S. average for 2023 [5] and the proposed German 2030 limit [9]).

2.3 SIPOC

Suppliers	Inputs	Process	Outputs	Customers
Electric utility; equipment vendors (UPS, generators, chillers); water utility; IT tenants; staff and contractors	Grid power; IT load; cooling water; procedures (SOP/MOP/EOP); maintenance	Receive power → condition and back up → power IT → remove heat → monitor → meter and report	Available IT capacity; PUE/WUE figures; outage log	Business users and tenants; regulators; community and investors

2.4 Lean waste observed in the process (documented)

Lean waste	Documented example	Source
Over-processing / conflicting controls	Cooling units “fighting” each other — some humidifying while others dehumidify — mean real PUE is generally worse than design	[5]
Idle capacity (inventory)	LBNL models conventional servers idling at 36% of maximum power in 2023, and internal data centers averaging about 20% utilization by 2027; only ~42% of non-colocation operators collect server-utilization data	[5] [1]
Defects / rework	47% of operators had an impactful outage in three years; 89% of impactful outages judged avoidable	[1]
Waiting / downtime	Outage cost (direct, opportunity, reputation) ≥ $100,000 for 71% of those with a significant outage	[1]
Under-utilized / scarce talent	53% report difficulty finding qualified staff; electrical skills gap 38%; 33% report staff hired away	[1]

3. MEASURE Phase

3.1 Operational definitions and data sources

CTQ	Operational definition	Source / sample	Data type
1 PUE	Total facility energy ÷ IT equipment energy, annual average for the respondent's data center (Green Grid metric, 2007)	Uptime 2026, n=644; Google fleet TTM PUE (measured) [1] [6]	Continuous
2 Impactful-outage incidence	Share of operators reporting ≥ 1 impactful outage in the past three years	Uptime 2020–2026, n=494–781 per year [1]	Binary / proportion
3 Outage preventability	Share of operators with an impactful outage who say it could have been avoided by better management, processes or configuration	Uptime 2024–2026; 2025 n=98 [2] [1]	Binary / proportion
4 Water-use tracking	Share of operators collecting water-usage data for sustainability purposes	Uptime 2025 (n=617), 2026 (n=593) [1]	Binary / proportion

Measurement-system notes. Operational definitions are those of the source surveys and were not altered. Limitations: responses are self-reported; the sample is not random (52% of respondents are in North America and Europe) [10]; PUE is best used to compare one site over time, not a few different facilities [1]; the LBNL PUE (1.4 in 2023) is *modelled* assuming systems run as designed and is therefore optimistic [5]; outage-cost samples are small (n=77 in 2026). Google's PUE is measured continuously for large-scale sites once stable [6].

3.2 Baseline performance

CTQ	n	Baseline (2026)	Spread / 95% CI	Meeting specification	Project target
PUE	644	1.52 (overhead 0.52)	SD not published	23% of sites < 1.3; share ≤ 1.5 not published	Average ≤ 1.40
Impactful-outage incidence	715	47%	43%–51% (computed)	53% report none; 10% of outages serious/severe	≤ 35%
Outage preventability	n/a	89% avoidable	Trend 80% → 87% → 89% (2024–26)	11% not avoidable	≤ 75%
Water-use tracking	593	53%	49%–57% (computed)	53% collect data	≥ 79%
(Context) Cost ≥ $100k	77	71%	61%–81% (computed)	29% < $100k; 19% > $1M	—

Measure conclusion. Every CTQ is short of its project target. The largest relative gaps are outage preventability (89% vs ≤ 75%) and water tracking (53% vs ≥ 79%). PUE is stagnant near 1.5 despite the industry's efforts; outage frequency is improving but consequences are rising (cost ≥ $100k rose from 57% to 71% in one year, though with n=77 that change has p = 0.06; computed).

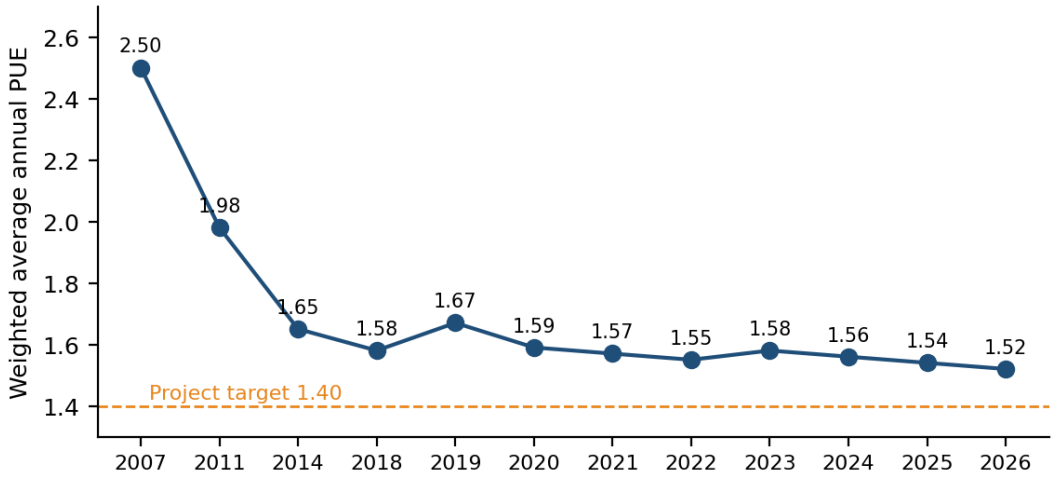


Figure 2. Uptime Institute weighted-average annual PUE, 2007–2026. Source: Uptime Global Data Center Surveys 2025 and 2026 [2] [1].

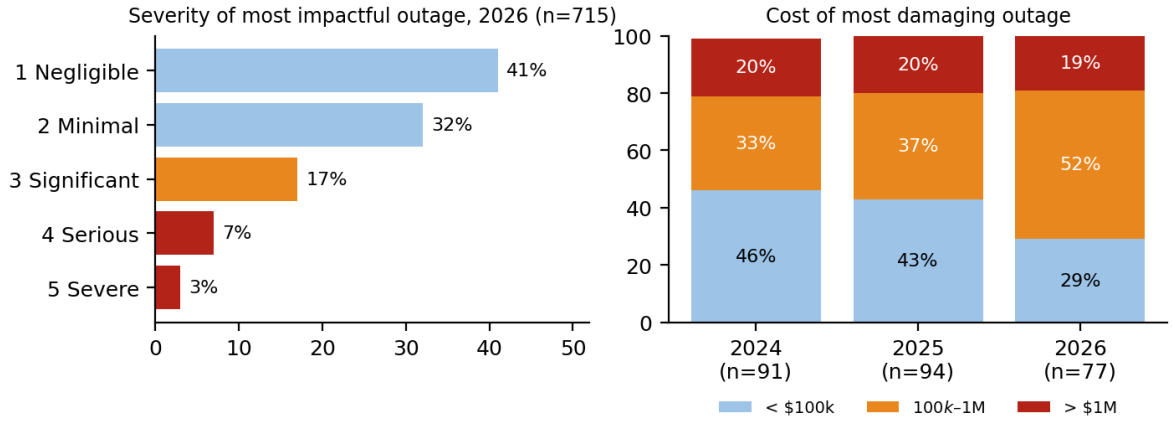


Figure 3. Left: severity of the most impactful outage (2026). Right: cost of the most damaging outage, 2024–2026 (only respondents with a significant, serious or severe outage). Sources: [1] [2].

4. ANALYZE Phase

4.1 Where do outages come from?

Power is the leading cause of impactful outages every year since 2020: 56% of respondents in 2026 (45% in 2025, 54% in 2024) [1] [2]. Within power-related IT outages, UPS failure is cited most often (39%), followed by transfer-switch (34%) and generator (28%) failures (n=129; multiple answers possible, so the shares are not additive and this is a ranked bar chart rather than a cumulative Pareto) [1]. Uptime cautions that root causes are hard to isolate and that the 2025 power share looks unusually low [1]. Externally, Uptime Intelligence reports rising connectivity/fibre and grid-related risk [11].

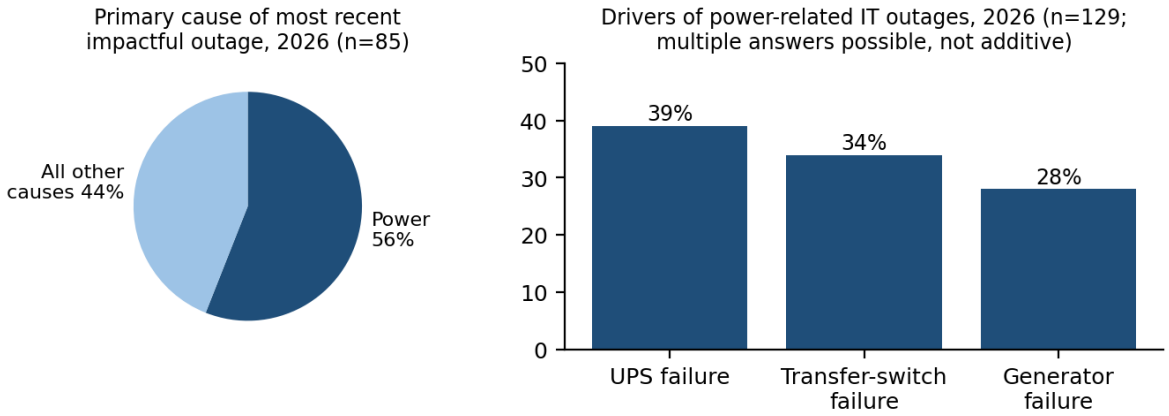


Figure 4. Primary cause of impactful outages (left) and drivers of power-related outages (right), 2026. Source: [1].

4.2 Cause-and-effect analysis

A fishbone diagram organised possible causes before deciding which were supported by data. Each entry is a published figure.

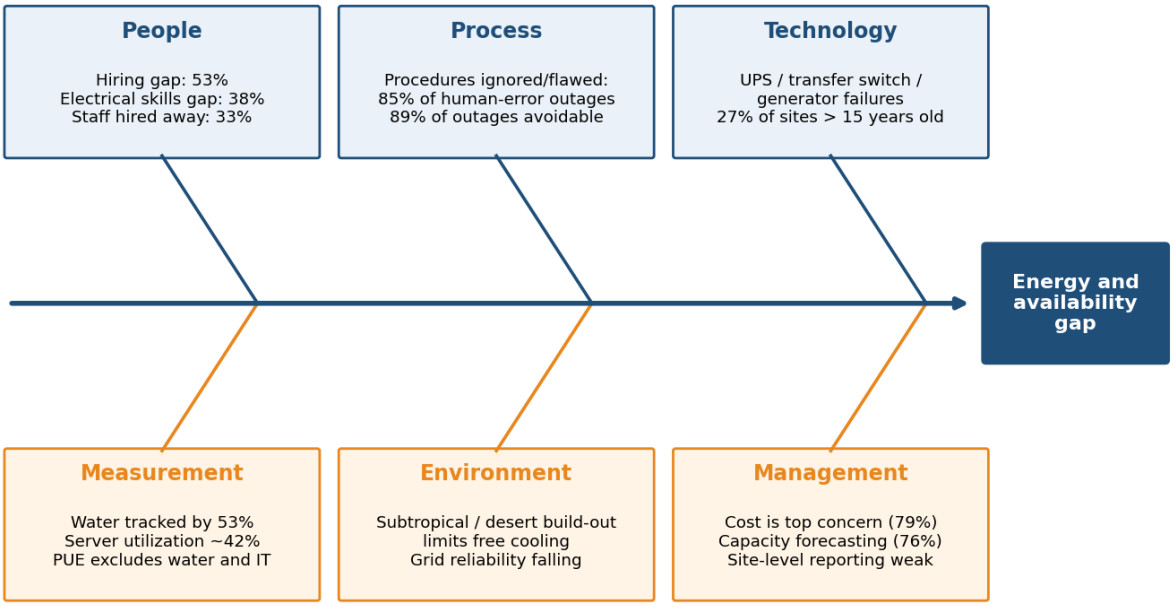


Figure 5. Fishbone analysis. Sources: [1] [3].

4.3 Root-cause verification (stratified comparisons of published data)

Suspected cause	Published comparison	Evidence	Source
Legacy / older facilities	Sites commissioned within ~5 years: PUE 1.48 vs all respondents 1.54 (2025); 27% of respondents work in sites > 15 years old (2026)	−0.06 PUE for newer sites	[2] [1]
Small scale	Data centers ≥ 20 MW: 1.44 vs all 1.54 (2025). Largest site 1.45 vs all 1.52 (2026)	−0.10 / −0.07 PUE for larger sites	[2] [1]
Hot / humid climate	Google Singapore campuses (Q2 2026 TTM): 1.12 and 1.15 vs fleet 1.09 — even best-in-class pays a tropical penalty; Uptime expects build-out in subtropical/desert regions to slow PUE gains	+0.03 to +0.06 PUE vs fleet	[6] [1]
Seasonal cooling load	Google fleet quarterly PUE, 2019Q1–2026Q2 (30 quarters): Q3 mean 1.120 vs Q4 mean 1.084 (computed)	+0.036 in the hottest quarter	[6]
Uncontrolled cooling / control	ML cooling control cut cooling energy 40% and PUE overhead 15% at a Google site; LBNL notes real systems rarely meet design	−15% overhead	[7] [5]
Procedures not followed or flawed	~40% of organizations had a major human-error outage in 3 years; 85% of these stem from ignored or flawed procedures; failure to follow procedures remains the leading driver in 2026	85% of human-error outages	[3] [4]
Power-chain equipment	Power = 56% of primary causes; UPS 39%, transfer switch 34%, generator 28% of power-related outages	56%	[1]
Skills and staffing	53% report difficulty finding qualified candidates; electrical skills gap 38%	53% / 38%	[1]
Water outside KPI set	Only 53% collect water data vs 79% for PUE; low-PUE evaporative cooling can raise water use	26-point gap	[1] [5]

4.4 Statistical tests on the published series (computed)

Test	Result	Interpretation
I-MR chart, industry PUE 2019–2026 (8 points)	CL 1.573; $\overline{MR}$ 0.030; UCL 1.652; LCL 1.493. 2026 (1.52) is inside the limits	Process is statistically stable: change is incremental, not a step. Regression 2020–2026 slope −0.0093/yr → ~13 years to reach 1.40 at that pace
p-chart, outage incidence 2020–2026	$\bar{p}$ = 57.5%. 2020–2022 above UCL; 2025 and 2026 below LCL	A real, sustained improvement in outage frequency
Two-proportion z, 2020 vs 2026 (74% vs 47%)	$z = 9.35$, $p < 0.001$	Highly significant improvement
Two-proportion z, 2025 vs 2026 (50% vs 47%)	$z = 1.15$, $p = 0.25$	Not significant — pace of improvement has slowed
Two-proportion z, cost ≥ $100k 2025 vs 2026 (57% vs 71%)	$z = 1.89$, $p = 0.059$	Not significant at 0.05 (n=94 vs 77); watch the trend
Two-proportion z, water tracking 2025 vs 2026 (47% vs 53%)	$z = 2.09$, $p = 0.037$	Significant increase, yet still a minority-plus of operators

These tests treat each annual survey as an independent sample; respondents are not the same panel each year, so p-values are indicative.

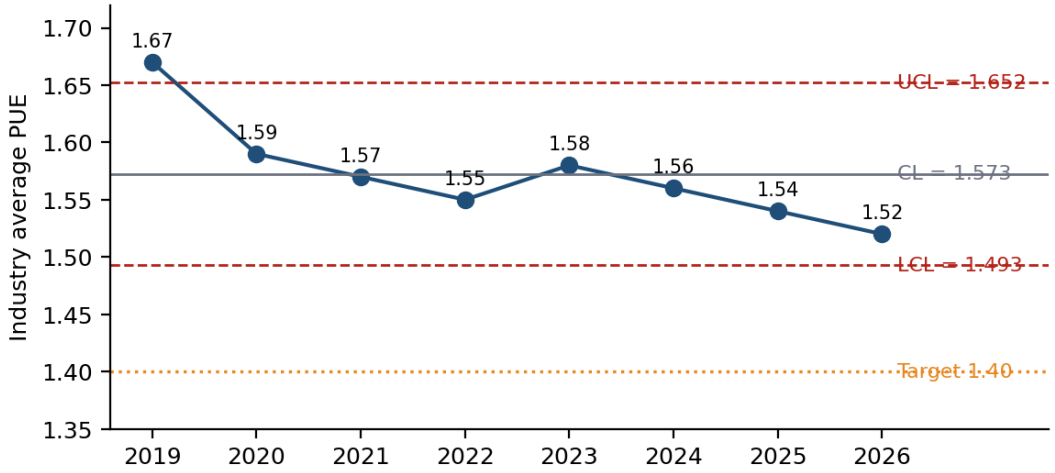


Figure 6. Individuals chart of published industry-average PUE, 2019–2026 (limits computed by the author from the published points; 1.40 is the project target). Source data: [2] [1].

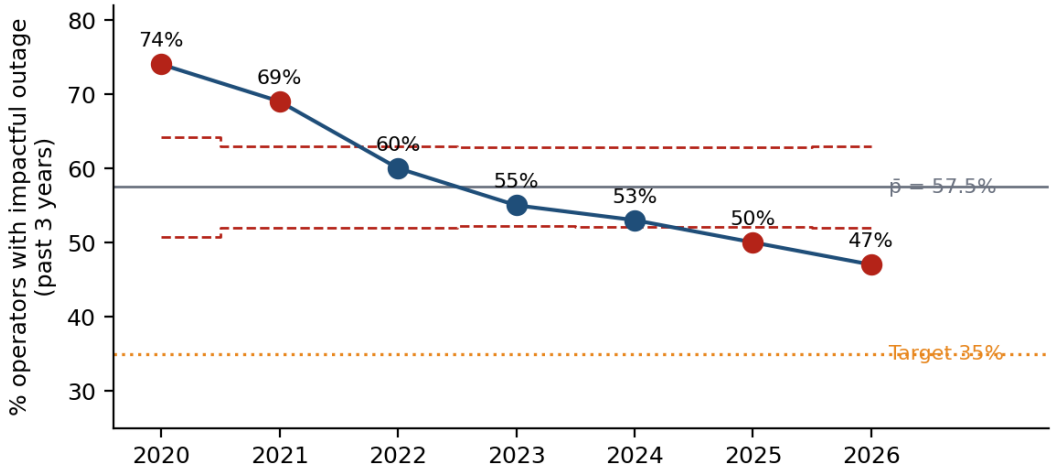


Figure 7. p-chart of the share of operators with an impactful outage in the past three years (red = outside 3-sigma limits; limits use each year's n). Source data: [1].

4.5 Five-Whys root-cause statements

The chains below are the author's synthesis of the cited evidence, not measured facts; each link is anchored to a source.

Problem	Why-chain (summary)	Verified root-cause statement
PUE stuck near 1.5	Average barely moves → cheap gains (containment, setpoints, controls) already taken [1] → deeper upgrades in live legacy sites are costly/disruptive [1] → 27% of sites are > 15 years old and kept for capacity [1] → no site-level efficiency roadmap	Easy efficiency gains are exhausted and legacy sites stay in service without a site-level retrofit or retirement plan.
Outages still frequent	Power chain fails (56%; UPS, switch, generator) [1] → equipment not tested/maintained/operated as configured [1] → 85% of human-error outages involve ignored or flawed procedures [3] → skills shortage (53%) [1] → no standard work and change control for the power chain	Weak procedural discipline and insufficient testing on the power chain, compounded by skills shortages.
Preventability rising (80→89%)	Redundancy invested (44% 2N, 40% N+1) [1] → value depends on configuration, maintenance, testing and operation [1] → physical redundancy not paired with operational discipline	Hardware redundancy is being bought without matching process and training investment.

Problem	Why-chain (summary)	Verified root-cause statement
Water blind spot	47% do not track water [1] → tracking driven by regulation/pressure, not by operations [1] [2] → PUE-only KPI culture → PUE optimisation can shift consumption to water [5]	Water use is not part of the standard facility KPI set, so trade-offs are invisible.

4.6 Risk prioritisation (by published frequency)

Failure mode	Published frequency	Main effect	Priority action
Power-chain failure	56% of primary outage causes; UPS 39%, transfer switch 34%, generator 28% of power outages	Outage, cost ≥ $100k (71%)	Power-chain maintenance and test programme
Procedure not followed / flawed	~40% of organizations; 85% of human-error outages	Avoidable outage (89%)	Standard work, checklists, change control, training
Cooling overhead above achievable	Overhead 0.52 vs 0.09 at Google (83% less)	Energy cost, regulatory exposure	Control optimisation, containment, retrofit/retire
Water use unmeasured	47% of operators do not track	Hidden risk in water-stressed sites	Meter and report WUE per site

5. IMPROVE Phase

5.1 Selected countermeasures

Verified root cause	Countermeasure	Published evidence	LSS rationale	Owner role
Easy gains exhausted; legacy sites	Airflow management, aisle containment, blanking panels, higher supply temperature, optimised cooling controls; retire or migrate worst sites	These measures drove PUE from 2.50 (2007) to 1.65 (2014) [2]; new builds routinely reach ≤ 1.3 [1]	Standard work; remove over-processing	Facility Engineering Mgr
Cooling controlled by fixed rules	ML/analytics recommendations for cooling setpoints, starting in advisory mode	−40% cooling energy, −15% PUE overhead at a Google site [7]; 71% of operators trust AI for analytics but only 31% for setpoint control [1]	Reduce variation; data-driven control	Facility Engineering Mgr
Power-chain failures	Preventive maintenance and periodic load/transfer tests of UPS, transfer switches, generators	These three account for 39%/34%/28% of power-related outages [1]; testing and training are the cheapest, most effective resilience measures [2]	Defect prevention	Operations Mgr
Procedures ignored or flawed	Maintain SOP/MOP/EOP library, checklists, change-control gates, post-incident reviews	85% of human-error outages involve procedures [3]; failure to follow procedures remains the leading driver [4]	Standard work; poka-yoke	Procedures Owner
Skills shortage	Structured training; partnerships with trade schools and colleges	Uptime recommends trade-school partnerships for electrical roles [1]	Develop under-used talent	Procedures Owner
Water outside KPI set	Meter water per site, report WUE beside PUE, weigh PUE–WUE trade-off when choosing cooling	LBNL: modelled U.S. average site WUE ≈ 0.36 L/kWh; hyperscale sites report 0.1–0.3; waterless chillers cost more energy [5]	Visual management; measure what matters	Sustainability Mgr

Relevance to tropical operators. Uptime expects data-center development in subtropical and desert regions to temper industry PUE gains and points to elevated-temperature liquid cooling as a longer-term help [1]. Google's Singapore campuses run at 1.12–1.15 TTM PUE [6] — evidence that ≤ 1.15 is achievable in the tropics at hyperscale, though not necessarily for small enterprise sites.

5.2 Implementation plan (proposed 12-week cycle)

Weeks	Phase	Key activities
1–2	Define	Confirm charter, CTQs, owners; agree data sources
3–4	Measure	Collect 12 months of site PUE/WUE, outage log, procedure-compliance audit; validate meters
5–7	Analyze	Site-level I-MR charts; Pareto of own outage/near-miss causes; verify root causes on own data
8–10	Improve	Implement containment/controls, power-chain test plan, standard-work library; run advisory-mode optimisation
11–12	Control	Install charts, owners, reaction plans; hand over to process owners

This is a plan for an operator adopting the method; it has not been executed at a real facility, so no measured "after" results are claimed.

5.3 Evidence of achievable performance (in place of a pilot)

CTQ	Baseline	Published evidence of what is achievable	Assessment vs target
PUE	1.52	Google fleet 1.09 (2025 TTM); Uptime largest sites 1.45; LBNL modelled U.S. 1.4; 23% of sites already < 1.3 [6] [1] [5]	1.40 is demonstrably achievable; hyperscale values reflect scale and design that legacy sites cannot copy
Outage incidence	47%	Fell 74% → 47% in six years, average −4.5 points/yr (regression, computed) [1]	35% needs ~3–4 more years at the recent 3-point pace ((47−35)/3 = 4.0 yrs) — a targeted programme should beat the trend
Preventability	89%	Rising 80% → 87% → 89%; no published improvement	Target ≤ 75% requires deliberate process discipline; no benchmark yet
Water tracking	53%	Rose 47% → 53% in one year (p = 0.037, computed) [1]	79% needs ~4 years at +6 points/yr ((79−53)/6 = 4.3 yrs); regulation (EU EED) is accelerating it

5.4 Projection: how much PUE change is needed? (arithmetic, not a measured result)

PUE = 1 + overhead. Baseline overhead is 0.52. If the DeepMind result (−15% overhead) were achieved industry-wide, PUE = 1 + 0.52 × 0.85 = **1.442**. To reach 1.40, overhead must fall to 0.40, i.e. a **23.1%** reduction. Google's fleet overhead (0.09) is 83% below the industry average [6]. Conclusion: control optimisation alone is not enough to reach 1.40; it must be combined with retrofit or retirement of the least efficient sites.

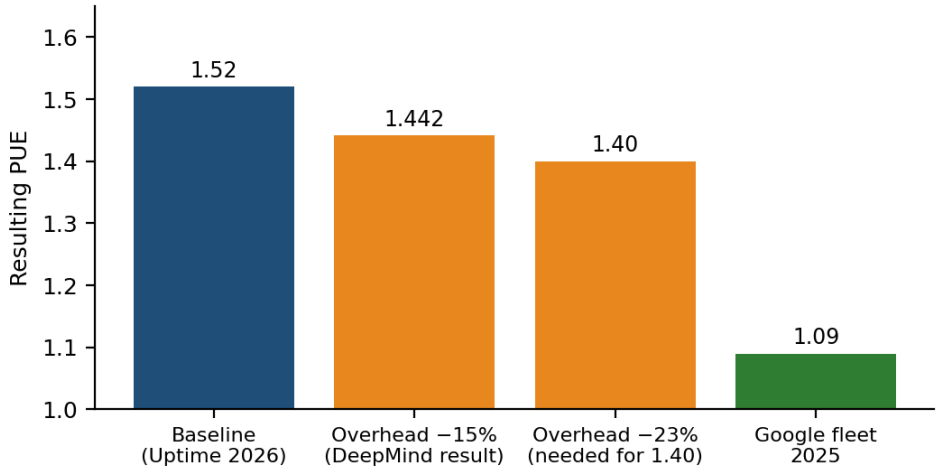


Figure 8. Projection of PUE under overhead reductions (computed from published inputs [1] [7] [6]).

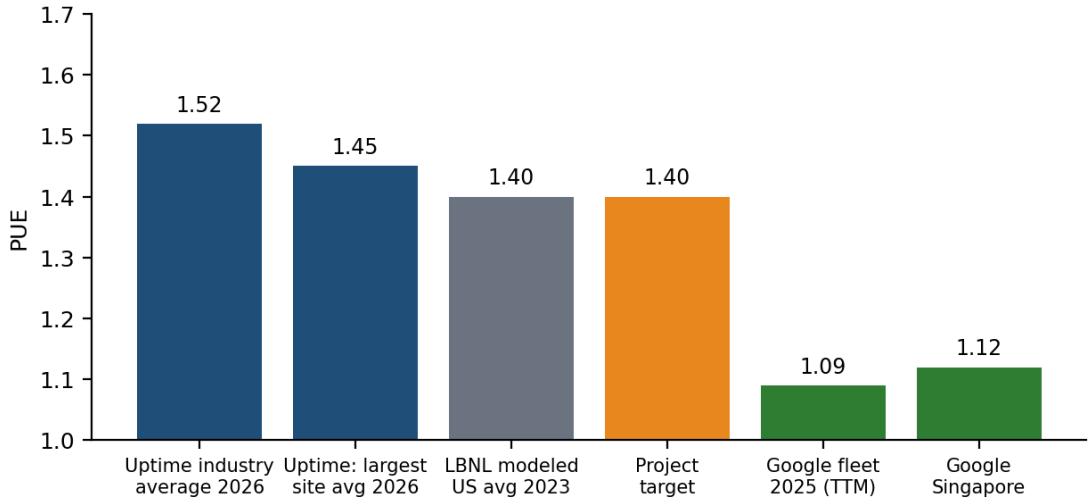


Figure 9. Benchmarks used to set the PUE target. Definitions differ (survey self-report, LBNL model, Google measurement), so bars are context, not a like-for-like league table. Sources: [1] [5] [6].

6. CONTROL Phase

6.1 Control plan

CTQ	Control standard	Chart / frequency	Owner	Reaction plan
PUE	Site TTM PUE ≤ 1.5 (individual); portfolio average ≤ 1.40; monthly PUE reported to management	I-MR chart on monthly PUE, limits computed from the site's own first 12 months; TTM value reported (Google practice)	Facility Engineering Mgr	Point outside limits, or 8 in a row above the centre line: review cooling controls, containment, setpoints and metering within 5 working days
Impactful-outage incidence	Rolling 36-month share of sites with an impactful outage ≤ 35%; zero serious/severe outages	p-chart, quarterly; every outage logged with cause category	Operations Mgr	Any impactful outage: root-cause review; any serious/severe: executive review and procedure audit
Outage preventability	≤ 75% of impactful outages classified avoidable; 100% of power-chain tests completed on schedule	Monthly count of tests done vs planned; classification of each outage	Procedures Owner	Missed test or procedure deviation: corrective action and retraining before next change window
Water tracking	≥ 79% of sites metering water and reporting WUE annually; 100% of sites in water-stressed areas	Quarterly coverage check; WUE run chart	Sustainability Mgr	Missing meter or WUE data at a site: install/repair plan within one quarter

Numeric limits for an operator's own site must come from that site's data. The industry limits shown in Figure 6 (1.493–1.652) demonstrate the method only.

6.2 Standardisation and handoff

- Cooling-control and containment standards (setpoints, allowed ranges) with change approval.
- Power-chain maintenance and test schedule; SOP/MOP/EOP library with owners and review dates.
- Change-management checklist for any work on live power or cooling paths.
- Outage log with cause categories and "avoidable?" classification for each incident.
- Site metering standard (power, IT load, water) and monthly PUE/WUE dashboard reviewed by department heads.

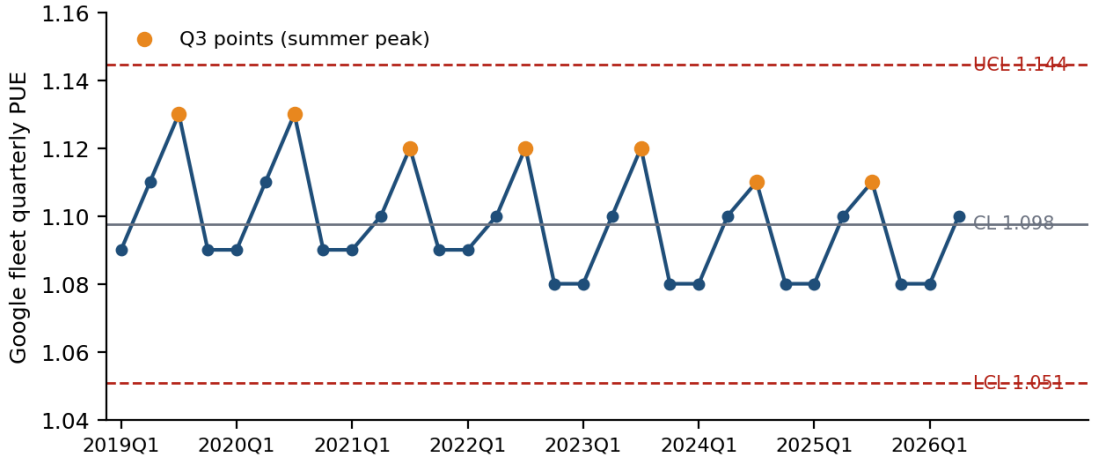


Figure 10. Real example of a control chart on measured PUE: Google fleet quarterly PUE, 2019Q1–2026Q2. All 30 points lie inside computed limits (1.051–1.144) but the summer quarter (orange) is systematically higher — control charts on PUE should account for seasonality or use trailing-twelve-month values. Source data: [6].

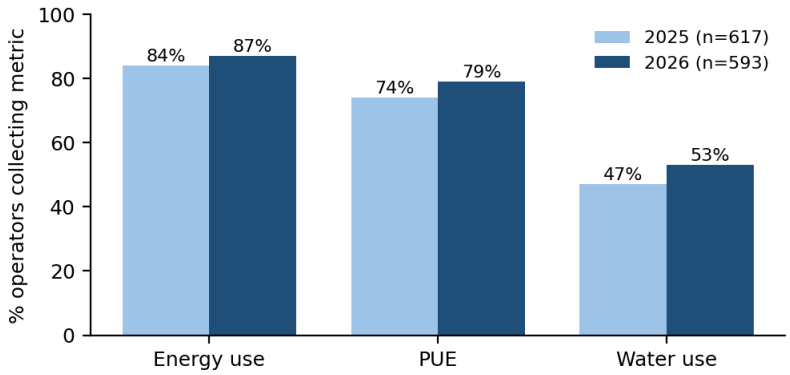


Figure 11. Share of operators collecting each metric, 2025 vs 2026. Source: [1].

6.3 Project closure criteria

The project may be handed back to process owners after two consecutive quarters in which (a) no unresolved special-cause signal appears on the PUE I-MR or outage p-charts, (b) all four CTQs are at or near their control standards, (c) the standard-work documents are in use, and (d) named owners have accepted responsibility for monitoring and corrective action.

7. Operational Impact

No financial savings are claimed, because a verified local electricity tariff, water price and outage-cost record are needed. Operational effects can be computed directly from the published numbers.

Impact area	Basis	Estimated operational effect (computed)
Energy per MW of IT load	1 MW × 8,760 h = 8,760 MWh IT energy. Facility energy = IT × PUE.	At PUE 1.52: 13,315 MWh/yr; at 1.40: 12,264 MWh/yr → **1,051 MWh/yr less (7.9%)**. At PUE 1.442 (−15% overhead): 683 MWh/yr less.
Overhead energy	PUE 1.52 → 1.40 per kW of IT	0.52 → 0.40 kWh of overhead per kWh of IT (−23%)
Outage exposure	71% of significant outages cost ≥ $100k; 19% > $1M [1]	Each avoided impactful outage avoids a cost in these bands; at 47% → 35% incidence, about 12 fewer operators in 100 affected per three years
Water visibility	Tracking 53% → 79%	26 more sites in 100 able to manage water risk and report WUE
System context	U.S. data-center electricity 176 TWh (2023); infrastructure (non-IT) was 30% of the total in 2023 [5]	Efficiency in cooling and power distribution addresses the 30% non-IT share

To convert to money, use the site's verified tariff, actual water costs, and actual outage cost records; keep estimated and verified savings separate.

8. Conclusion

DMAIC gives a disciplined way to turn published data-center performance data into an improvement plan. Define translated stakeholder concerns into four CTQs; Measure fixed baselines from the latest Uptime survey (PUE 1.52; 47% impactful-outage incidence; 89% avoidable; 53% water tracking) [1]. Analyze showed that PUE is a stable process improving by less than 0.01 per year, that outage incidence has genuinely fallen but recently stalled, and that power-chain equipment and procedural discipline dominate outage causes. Improve linked each verified cause to a countermeasure with published results and showed by arithmetic that machine-learning cooling control alone (PUE ≈ 1.44) would not reach the 1.40 target. Control specified charts, owners and reaction plans. The main limitation is that all data are published aggregates: an operator adopting this plan should re-run the Measure and Analyze steps on its own site data before committing to targets.

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All sources were accessed between 28 and 30 September 2026. Click the URL to open each source.

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Appendix A — Data extract used (all from published sources)

A1. Uptime industry-average annual PUE [2] [1]

Year	2007	2011	2014	2018	2019	2020	2021	2022	2023	2024	2025	2026
PUE	2.50	1.98	1.65	1.58	1.67	1.59	1.57	1.55	1.58	1.56	1.54	1.52

A2. Share of operators with an impactful outage in the past three years [1]

Year	2020	2021	2022	2023	2024	2025	2026
Share	74%	69%	60%	55%	53%	50%	47%
n	494	730	730	781	768	754	715

A3. Cost of most damaging outage (respondents with a significant, serious or severe outage) [2] [1]

Year (n)	< $100k	$100k–$1M	> $1M
2024 (91)	46%	33%	20%
2025 (94)	43%	37%	20%
2026 (77)	29%	52%	19%

Percentages are as printed in the sources and may not sum to 100 because of rounding.

A4. Google fleet quarterly PUE [6]

Year	Q1	Q2	Q3	Q4
2019	1.09	1.11	1.13	1.09
2020	1.09	1.11	1.13	1.09
2021	1.09	1.10	1.12	1.09
2022	1.09	1.10	1.12	1.08
2023	1.08	1.10	1.12	1.08
2024	1.08	1.10	1.11	1.08
2025	1.08	1.10	1.11	1.08
2026	1.08	1.10	—	—

Appendix B — Calculation examples

Measure	Formula	Calculation
PUE overhead	PUE − 1	1.52 − 1 = 0.52
Overhead cut needed for 1.40	1 − (0.40 ÷ 0.52)	1 − 0.769 = 23.1%
PUE after −15% overhead	1 + 0.52 × 0.85	1.442
Individuals chart limits	CL ± 2.66 × $\overline{MR}$	1.573 ± 2.66 × 0.030 → 1.493 to 1.652
p-chart limits	$\bar{p} \pm 3\sqrt{(\bar{p}(1-\bar{p})/n)}$	$\bar{p}$ = 0.5746 (n-weighted mean of 7 yearly proportions); 2026 limits 0.519–0.630
95% CI for a proportion	$p \pm 1.96\sqrt{(p(1-p)/n)}$	Outage 2026: 0.47 ± 0.037 → 43%–51%
Two-proportion z	$(p1 - p2) \div \sqrt{(\hat{p}(1-\hat{p})(1/n1+1/n2))}$	2025→2026 outages: z = 1.15
Outage-reduction requirement	(1 − 35/47) ÷ 0.89	28.7% of avoidable outages must be eliminated to reach 35%
Energy per MW IT	8,760 h × PUE	8,760 × 1.52 = 13,315 MWh; × 1.40 = 12,264 MWh

Appendix C — Limitations and data-integrity statement

- All inputs are published figures; the author's own work is arithmetic, charts and synthesis, each marked "computed" or "synthesis".
- Uptime figures are survey self-reports; sample composition and question wording change slightly by year. Percent-of-operators is not percent-of-sites.
- Uptime's 2025 survey placed water tracking at 47%; the 2026 figure of 53% was read from Figure 6 of the 2026 report, whose executive text states only that "more than half" now track water.
- The outage-cause breakdown beyond "power" could not be mapped unambiguously from the extracted figure and is therefore not used.
- LBNL PUE/WUE values are modelled; Google values are measured for its own large-scale sites; neither represents small enterprise rooms.
- The German PUE rules were current law at the time of the White & Case summary; a 2026 draft bill would relax them and had not been confirmed at the time of the Gleiss Lutz note.
- No pilot was run; "Improve" results are published evidence and projections, not measured outcomes.

Appendix D — Traceability of key numbers

Number	Where to find it
PUE 1.52 (n=644); 23% < 1.3; largest site 1.45; 27% > 15 yrs	[1] “PUE edges downward”, Figure 2
47% impactful outage (n=715); 10% serious/severe	[1] Figures 11–12
Cost: 29% / 52% / 19% (n=77)	[1] Figure 13
Power 56% (n=85); UPS 39%, transfer switch 34%, generator 28% (n=129)	[1] Figure 14 and text
89% avoidable (87% in 2025, 80% in 2024)	[1] “Building more resilient systems”
Energy 87%, PUE 79%, water 53% tracked	[1] “Sustainability and metrics”, Figure 6
~40% human-error outages; 85% procedures	[3]
−40% cooling energy; −15% PUE overhead	[7]
Google fleet PUE 1.09; quarterly series; Singapore 1.12/1.15	[6]
176 TWh (2023); 4.4%; 325–580 TWh (2028); avg PUE 1.4; WUE 0.36 L/kWh; infrastructure 30%	[5] [12]
German PUE limits 1.5 / 1.3 (existing), 1.2 (new); draft 1.6 / 1.4 / 1.3	[8] [9]